

Machine Learning-Based Dynamic Hosting Capacity Assessment for Electric Vehicles

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Abstract—This paper presents a measured-data framework for electric-vehicle (EV) hosting-capacity (HC) assessment in the IEEE European low-voltage distribution feeder. In contrast to synthetic constant-power EV assumptions, the proposed method replays measured grid-side active and reactive charging trajectories from an Opel Corsa-e experiment and augments the battery-state description using measured CALCE open-circuit-voltage–state-of-charge data. A stochastic core of 16,500 daily scenarios is generated by varying EV count, spatial placement, initial state of charge, and charging start time. Operational feasibility is evaluated using four hard constraints: voltage magnitude, line ampacity, distribution-transformer utilization, and daily network-loss rate. The unbiased core contains 1,490 feasible scenarios and yields a maximum observed aggregate feasible EV energy of $652.7386 \text{ kW h d}^{-1}$; low voltage is the binding constraint. A physics-valid enriched pool is then used for machine-learning (ML) classification and regression with 14 leakage-free scenario descriptors, including two charging-concurrency features. The best classifier is an RBF support-vector machine (SVM) with an F1-score of 0.9545, while support-vector regression achieves $R^2 = 0.9869$, $\text{MAE} = 6.5023 \text{ kW h}$, and $\text{RMSE} = 10.2364 \text{ kW h}$. Finally, a 55-location HC extension is performed with binary search under the same four constraints. Locational HC ranges from $120.81 \text{ kW h d}^{-1}$ to $1127.71 \text{ kW h d}^{-1}$, with a Spearman correlation of -0.9078 between feeder distance and HC. The results demonstrate that measured charging behavior, constraint-aware feasibility, data-driven prediction, and locational mapping provide complementary information for distribution-system planning.

Index Terms—Electric vehicles, hosting capacity, OpenDSS, Monte Carlo simulation, support vector machine, distribution network, state of charge, locational hosting capacity, machine learning.

I. INTRODUCTION

ELECTRIC vehicle integration introduces time-varying and spatially uncertain demand into low-voltage distribution systems. Hosting capacity (HC) provides a planning metric for quantifying the amount of new generation or load that can be connected without violating selected technical limits [6], [7]. For EVs, the result is strongly influenced by charging behavior, state of charge (SOC), connection time, location, and the operational limits adopted by the distribution system operator.

Karahan and Bagriyanik [1] proposed a two-stage stochastic HC and ML framework for bidirectional EVs on the IEEE European low-voltage test system. Their study generated 16,500 Monte Carlo scenarios, assessed voltage and overload indices by 24-h OpenDSS simulations, and trained classification and regression models to predict operational suitability and daily

EV power. The framework is attractive because physics-based simulations act as the source of training labels while ML is used as a fast surrogate for repeated planning queries. However, the EV behavior in that work is synthetic, based on an 80-kWh battery, fixed 3.7-kW charge/discharge power, and both G2V and V2G operation.

Recent HC literature also highlights the value of broader constraint sets and locational assessment. Xi *et al.* [2] formulate EV HC using voltage, current, transformer utilization, loss, and load-shape indicators. Qammar *et al.* [3] demonstrate that ML can learn feeder-specific HC and violated-constraint behavior from network descriptors. Node-level HC is particularly useful for planning because available capacity varies across connection points [4], [11]. Binary-search formulations provide an efficient way to estimate the maximum feasible capacity at each node [5].

The present work extends the stochastic HC/ML concept in four directions. First, measured Opel Corsa-e grid-side $P(t)$ and $Q(t)$ trajectories replace the fixed synthetic EV power assumption, and measured CALCE OCV–SOC data are introduced as battery-state descriptors. Second, feasibility is strengthened from voltage/line checks to a four-constraint formulation including transformer utilization and daily network loss. Third, ML is trained with a leakage-free 14-feature set that explicitly includes peak charging concurrency. Fourth, a separate 55-location HC map is obtained under the same physical limits to quantify spatial sensitivity. The main contributions are summarized as follows:

- A measured-data stochastic EV model with SOC-dependent remaining charging trajectories and measured grid-side active/reactive power.
- A four-constraint physical HC definition with an unbiased 16,500-scenario core retained as the sole source of natural feasibility and aggregate HC statistics.
- Leakage-controlled ML classification and regression using 14 scenario-side predictors, including explicit charging-concurrency descriptors.
- A bus-wise location-sensitivity extension using binary search and a common expected measured EV profile at all 55 load connection points.

II. MEASURED-DATA AND NETWORK MODELLING

A. Measured EV Charging and Battery-State Data

The EV input is derived from a measured Opel Corsa-e charging record containing 34,537 raw samples, of which 33,780 correspond to usable active charging. The measured

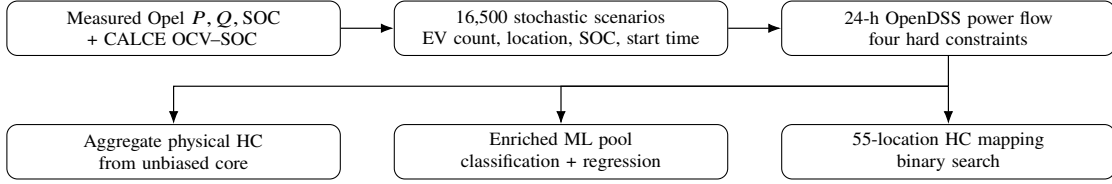


Fig. 1. Overall structure of the proposed measured-data EV hosting-capacity study.

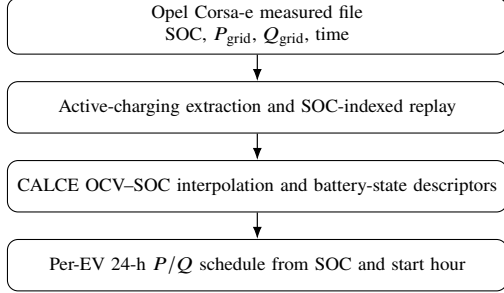


Fig. 2. Measured-data preprocessing and EV schedule construction.

active SOC support spans approximately 3–100%, the observed charging-session duration is 4.8276 h, and the measured grid-side energy is 46.6289 kWh. The median measured grid active power is 10.2082 kW and the median reactive-power magnitude is 1.9624 kvar. For stochastic simulations, the initial SOC is sampled over 3–99% so that a nonzero remaining charging trajectory is available.

The measured trajectory is replayed from the sampled initial SOC rather than imposing a fixed charger power. Thus, the EV demand at time t is obtained from the measured profile segment remaining after the initial SOC, preserving the measured taper behavior at high SOC. The CALCE dataset contributes 19 measured OCV points over approximately 3.2642–4.1758 V/cell, with usable cell capacity 1.9873 Ah. Mean and standard-deviation OCV descriptors are computed for each scenario from the sampled SOC values.

B. Distribution Feeder and Base-Day Audit

The IEEE European low-voltage feeder is modeled in OpenDSS with 55 load connection points and 907 OpenDSS buses. Public one-minute load profiles are converted to 24 hourly mean load shapes for the daily study. The no-EV base-day solution converges with minimum voltage 0.97149 pu, maximum voltage 0.999895 pu, maximum line loading 18.9646%, transformer utilization 5.1846%, and daily network-loss rate 0.8068%. These values establish a feasible base state under the final four constraints.

A peak-valley differential threshold of 85% was also examined because Xi *et al.* [2] use this quantity as a case-study flexibility indicator. However, the present no-EV feeder already has a peak-valley differential of 85.4959%. Consequently, transferring the 85% threshold directly would make the feeder infeasible even before EV integration. The peak-valley metric is therefore retained only as a supplementary indicator and is not used as a hard suitability constraint.

C. Stochastic Scenario Generation

The stochastic design follows the scenario structure of [1] while replacing synthetic EV power with measured charging. The EV-count levels are

$$N_{EV} \in \{5, 10, 15, \dots, 55\}, \quad (1)$$

with 1,500 independent scenarios at each level, producing 16,500 unbiased core scenarios. For every scenario, EV locations are sampled without replacement from the 55 load nodes, with at most one EV at each selected load node. Initial SOC values are sampled from 3–99%, and charging start hours are sampled from 0–23. Each EV then replays the remaining measured P/Q trajectory from its sampled SOC.

III. FOUR-CONSTRAINT HOSTING-CAPACITY FORMULATION

For scenario s , operational suitability is defined by the conjunction

$$\mathcal{S}^{(s)} = C_V^{(s)} \wedge C_I^{(s)} \wedge C_T^{(s)} \wedge C_L^{(s)}, \quad (2)$$

where the four hard constraints are

$$C_V : 0.95 \leq V_{n,h}^{(s)} \leq 1.05 \quad \forall n, h, \quad (3)$$

$$C_I : L_{\max}^{(s)} < 100\%, \quad (4)$$

$$C_T : U_T^{(s)} \leq 75\%, \quad (5)$$

$$C_L : \beta^{(s)} \leq 2.5\%. \quad (6)$$

The transformer-utilization and loss limits are adopted as literature-based case-study thresholds from [2]; they are not treated as universal grid standards. The daily network-loss rate is calculated as

$$\beta^{(s)} = 100 \frac{E_{\text{loss}}^{(s)}}{E_{\text{source}}^{(s)}}. \quad (7)$$

The aggregate feeder-level HC is defined from the unbiased stochastic core as the largest observed feasible daily EV energy,

$$HC_{\text{agg}} = \max_{s: \mathcal{S}^{(s)}=1} E_{EV}^{(s)}. \quad (8)$$

This definition intentionally keeps the physical HC separate from the enriched ML pool.

IV. MACHINE-LEARNING FORMULATION

A. Dataset Construction and Leakage Control

The unbiased core contains 1,490 suitable and 15,010 unsuitable scenarios under the final four-constraint rule. Because the suitable class is small, additional independent Monte Carlo proposals are evaluated with the same OpenDSS physics and unchanged constraints. Only physically suitable proposals are accepted until the suitable ML pool reaches 11,628 cases. The

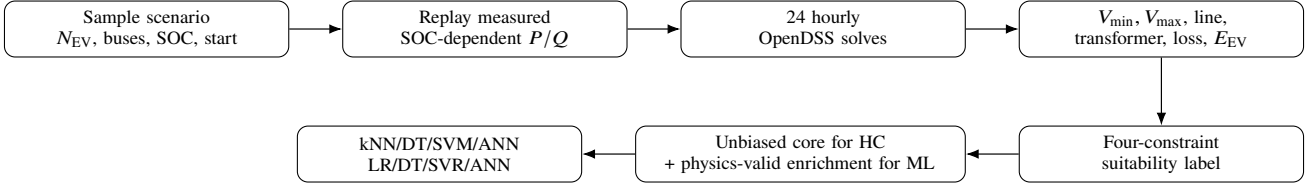


Fig. 3. Physics-to-ML workflow. The unbiased core is retained for physical HC; enrichment is used only for ML training/evaluation.

matched classification set contains 11,628 suitable and 4,872 unsuitable scenarios (16,500 rows), while regression uses all 11,628 suitable scenarios. The enriched/matched sets are used only for ML; they do not redefine the natural suitability rate or the physical HC in (8).

Direct constraint outputs—minimum voltage, maximum voltage, line loading, transformer utilization, network loss, and suitability labels—are excluded from the ML input vector to avoid target leakage. The final 14 predictors are

$$\mathbf{x} = [N_{EV}, d_\mu, d_\sigma, SOC_\mu, SOC_\sigma, t_\mu, t_\sigma, \bar{P}_0, \bar{Q}_0, \bar{PF}_0, \bar{OCV}, \sigma_{OCV}, N_{conc}^{pk}, r_{conc}]. \quad (9)$$

The two added concurrency descriptors are defined as

$$N_{conc}^{pk} = \max_h \sum_{i=1}^{N_{EV}} \mathbb{1}\{i \text{ is charging at hour } h\}, \quad (10)$$

$$r_{conc} = \frac{N_{conc}^{pk}}{N_{EV}}. \quad (11)$$

Thus, N_{conc}^{pk} measures the absolute worst-hour charging overlap, whereas r_{conc} measures the fraction of the scenario fleet simultaneously active at that peak. Both are derived only from sampled SOC/start-time behavior and the measured remaining charging duration; no power-flow output is used in their construction.

B. Models and Validation

Classification compares k-nearest neighbors (kNN), Decision Tree (DT), SVM, and artificial neural network (ANN) models. Regression compares linear regression (LR), DT, support-vector regression (SVR), and ANN. An 80/20 train-test split and 5-fold cross-validation are used. Inputs to SVM/SVR and other scale-sensitive models are standardized. Classification is evaluated with accuracy, precision, recall, and F1-score. Regression is evaluated with MAE, RMSE, and R^2 .

V. RESULTS AND DISCUSSION

A. Aggregate Physical HC and Constraint Behavior

The final 4C suitability count is 1,490 out of 16,500 (9.03%). The core contains 15,010 low-voltage violations, no high-voltage violations, 13 line-overload violations, no transformer-utilization violations, and 9,970 daily-loss violations. The additional transformer and loss constraints do not remove any of the 1,490 scenarios that were already feasible under voltage and line limits; therefore, voltage remains the dominant constraint for the aggregate HC.

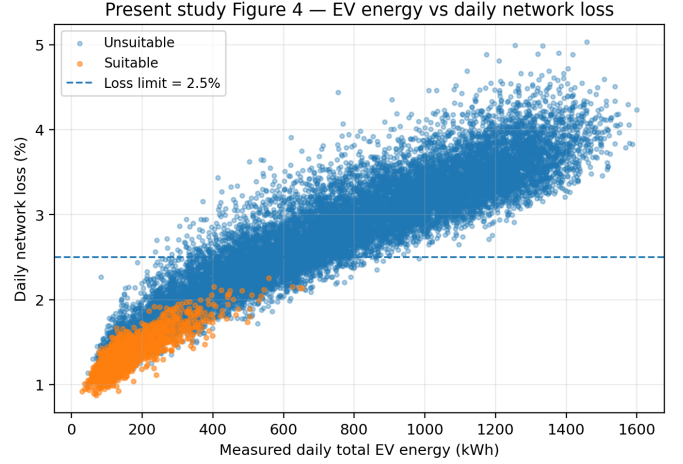


Fig. 4. Daily total measured EV energy versus daily network-loss rate for the unbiased 16,500-scenario core. The final loss limit is 2.5%.

The maximum observed feasible daily EV energy is

$$HC_{agg} = 652.7386 \text{ kWh d}^{-1}. \quad (12)$$

This occurs in Scenario 7487 with 25 EVs at unique randomly selected load nodes. At this point, $V_{min} = 0.951471$ pu, $V_{max} = 1.012007$ pu, maximum line loading is 44.9163%, transformer utilization is 11.5784%, and daily loss is 2.1355%. Hence, the lower-voltage boundary is closest to violation and is identified as the binding physical constraint.

The number 25 is not a manually chosen HC level. Feasible scenarios are observed at 5, 10, 15, 20, and 25 EVs; no feasible realization is found for 30 EVs or more. Specifically, 953, 390, 117, 24, and 6 feasible scenarios are obtained for 5, 10, 15, 20, and 25 EVs, respectively. The 652.7386-kWh/day value is the sum of the 25 individual measured-profile-based charging energies in the highest-energy feasible stochastic realization.

Fig. 4 shows the monotonic increase of network loss with aggregate charging energy. Although 9,970 core scenarios exceed the 2.5% loss threshold, all 1,490 final suitable cases remain below it; consequently, loss is an active screening constraint but does not reduce the maximum aggregate HC below the voltage-limited value.

As EV demand increases, minimum voltage decreases and eventually crosses 0.95 pu (Fig. 5). In contrast, maximum voltage remains below 1.05 pu for all scenarios (Fig. 6), which is consistent with the absence of measured V2G injection. The reference bidirectional framework in [1] includes both charging and discharging branches; the present G2V-only measured dataset therefore exhibits a different voltage-response envelope.

Fig. 7 summarizes the complete four-constraint feasibility view. Line loading and transformer utilization retain substantial

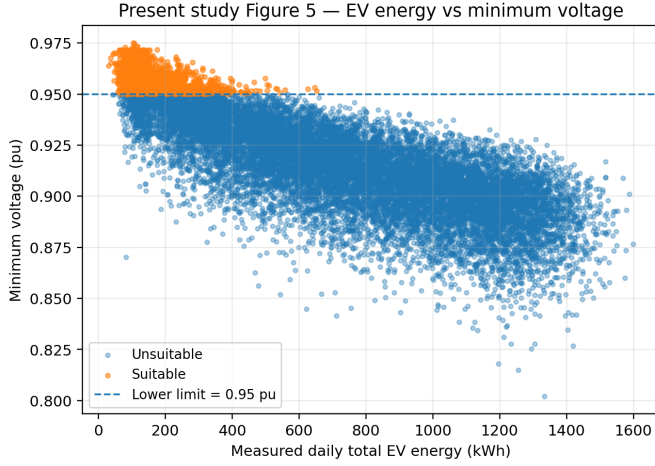


Fig. 5. Daily total measured EV energy versus minimum bus voltage. The 0.95-pu lower limit is the dominant feasibility boundary.

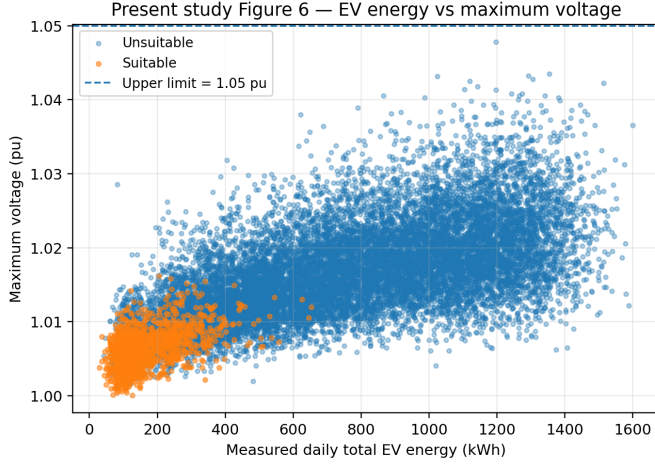


Fig. 6. Daily total measured EV energy versus maximum bus voltage. No high-voltage violation occurs because the measured dataset contains G2V charging only.

headroom at the HC boundary, while minimum voltage approaches 0.95 pu. Daily loss remains below 2.5% for all suitable cases. These results show that adding multiple constraints does not imply that every constraint must become binding; instead, the expanded formulation identifies which network limit is actually active for the studied feeder and operating regime.

B. Classification Results

Table I reports the final 14-feature classification results. SVM provides the highest test F1-score, 0.9545, with recall of 0.9776. The class distribution is acceptance-sampled for ML and therefore these metrics quantify discrimination on the matched ML set rather than the natural 9.03% suitable prevalence of the unbiased physical core.

The two concurrency features give only a marginal numerical improvement over the previous 12-feature model: the best F1-score changes from 0.954221 to 0.954460. Their main value is physical representation: EV count and start-time statistics alone do not explicitly state how many charging sessions overlap at the most stressful hour.

TABLE I
CLASSIFICATION PERFORMANCE WITH 14 LEAKAGE-FREE FEATURES

Model	Accuracy	Precision	Recall	F1
kNN	0.9258	0.9319	0.9652	0.9483
DT	0.9294	0.9366	0.9652	0.9507
SVM	0.9342	0.9323	0.9776	0.9545
ANN	0.9324	0.9361	0.9703	0.9529

TABLE II
REGRESSION PERFORMANCE FOR DAILY EV ENERGY PREDICTION

Model	MAE (kWh)	RMSE (kWh)	R^2
LR	12.2476	17.2304	0.9630
DT	8.2007	12.3961	0.9808
SVM	6.5023	10.2364	0.9869
ANN	7.8359	10.6394	0.9859

TABLE III
SELECTED BEST REGRESSION CONFIGURATION

Component	Final setting
Selected model	SVM/SVR with RBF kernel
Train-test split	80% / 20%
Validation	5-fold cross-validation
Feature set	14 leakage-free predictors
New feature 1	Peak_Concurrent_EVs
New feature 2	Concurrency_Ratio
C	100
ϵ	0.1
γ	0.01

TABLE IV
AGGREGATE HC SCENARIO AT THE FINAL FOUR-CONSTRAINT BOUNDARY

Metric	Value
Scenario ID	7487
EV count	25
Aggregate EV energy	652.7386 kWh/day
Minimum voltage	0.951471 pu
Maximum voltage	1.012007 pu
Max. line loading	44.9163%
Transformer utilization	11.5784%
Daily network loss	2.1355%
Binding constraint	Low voltage

C. Regression Results

Table II compares the four regression models. SVM/SVR produces the lowest MAE and the highest R^2 . ANN also performs strongly, but the measured G2V-only feature-target relationship is represented slightly better by the RBF-SVM in this dataset. Model selection is therefore data-driven rather than forced to match the ANN selected in [1].

The prediction points in Fig. 8 remain close to the ideal line over most of the target range. Compared with the previous 12-feature version, R^2 increases from 0.986661 to 0.986928, MAE decreases from 6.576876 to 6.502263 kWh, and RMSE decreases from 10.340346 to 10.236383 kWh. The improvement is intentionally described as marginal rather than substantial.

Present study Figure 7 — Final 4C performance indices for suitable scenarios

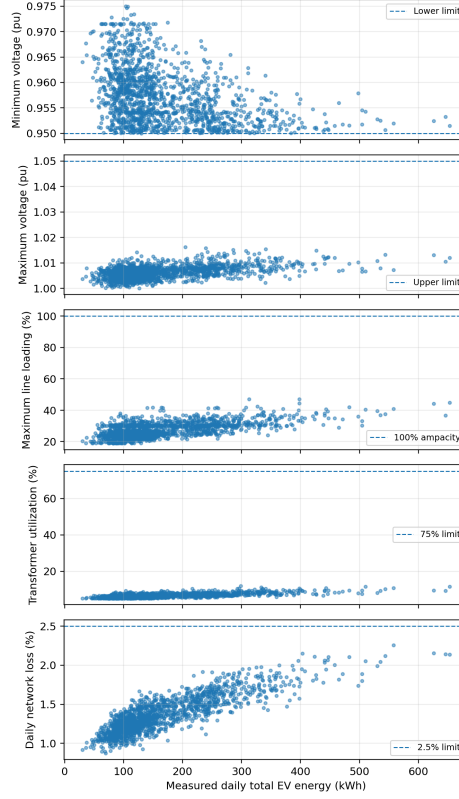


Fig. 7. Performance indices for the final 4C-suitable scenarios: minimum voltage, maximum voltage, line loading, transformer utilization, and daily network-loss rate.

14-feature regression model: SVM

Figure 8. Predicted vs actual — 14-feature model (SVM)

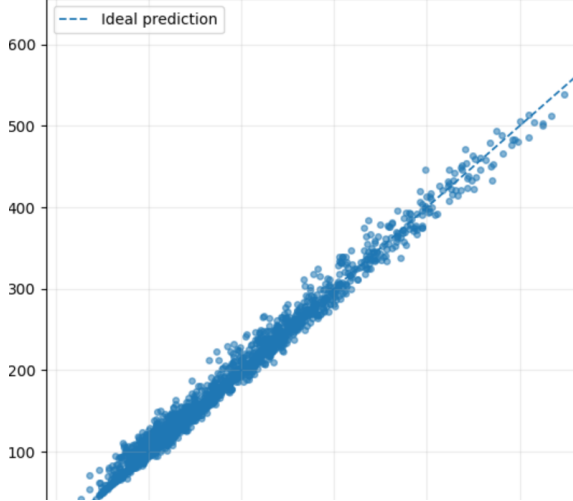


Fig. 8. Predicted versus actual measured daily EV energy for the selected SVM/SVR regressor.

VI. LOCATIONAL / BUS-WISE EV HOSTING CAPACITY

A. Definition and Search Procedure

The aggregate HC in (8) answers a stochastic multi-location question. A second analysis is therefore introduced to isolate location sensitivity. For this purpose, a common expected measured EV charging profile is formed by averaging all 2,328

combinations of integer initial SOC from 3–99% and start hour from 0–23. The resulting expected energy per EV-equivalent is 23.644101 kWh/day.

For each load location i , the same expected profile is scaled by a continuous factor α_i and applied only at that connection point. The locational HC is

$$HC_i = \max_{\alpha_i \geq 0} \alpha_i E_{EV, \exp} \quad \text{s.t.} \quad C_V \wedge C_I \wedge C_T \wedge C_L. \quad (13)$$

A binary search with 0.25-kWh/day energy tolerance is used to locate the maximum feasible value, following the efficient node-wise search concept used in [5]. The same four constraints are applied at every location so that the changing variable is location rather than charging behavior.

The quantities HC_i are separate counterfactual single-location capacities and must not be summed. In particular, $\sum_i HC_i$ is not equal to the aggregate stochastic HC of 652.7386 kWh/day because every HC_i reuses the full feeder capacity under a different hypothetical placement.

B. Locational Results

The strongest location is bus 34.1 with

$$HC_{34.1} = 1127.7128 \text{ kWh/day}, \quad (14)$$

corresponding to 47.695 EV-equivalents of the common expected profile. The limiting metric is network loss. The weakest location is bus 899.2 with

$$HC_{899.2} = 120.8066 \text{ kWh/day}, \quad (15)$$

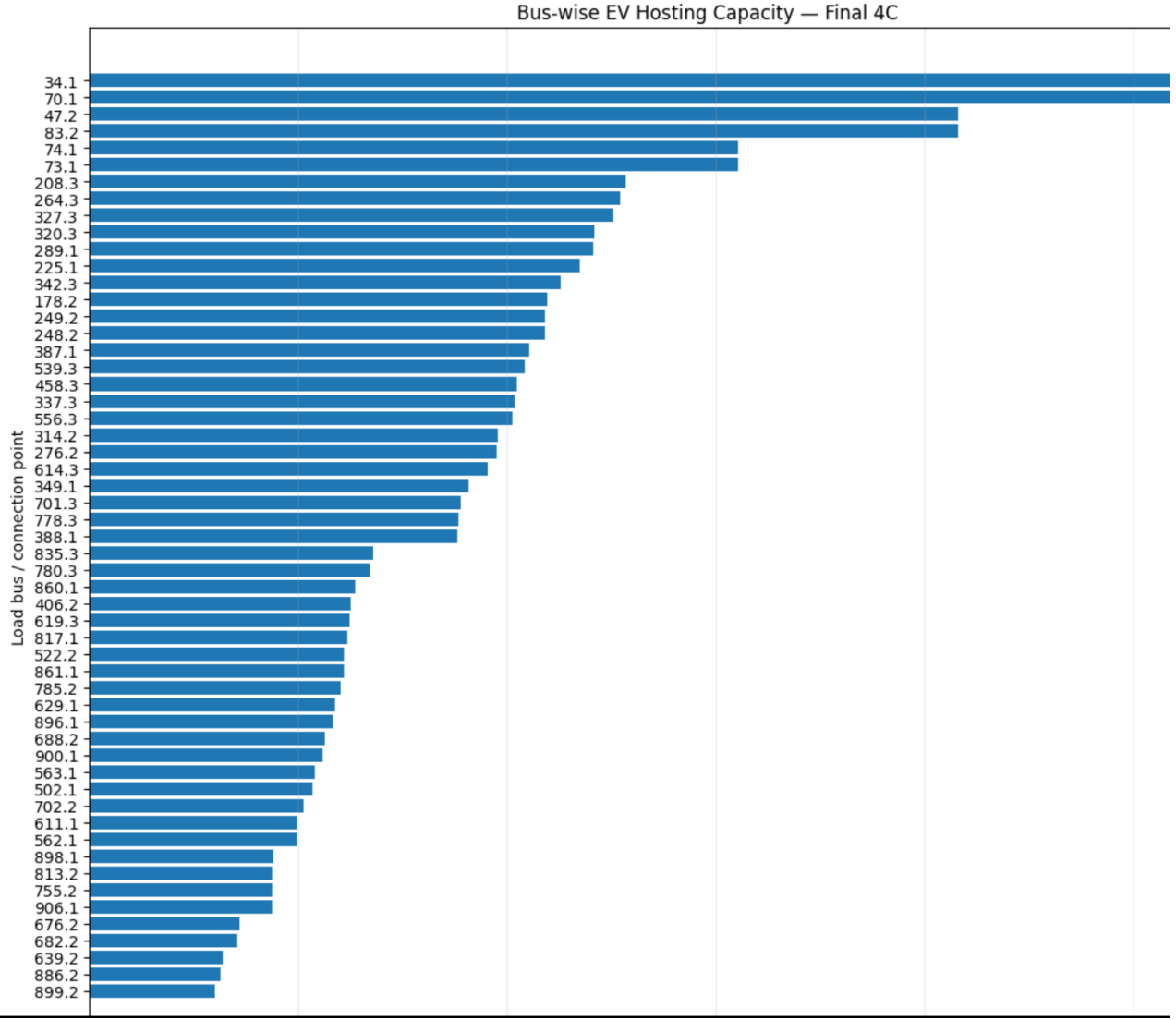


Fig. 9. Ranked locational EV hosting capacity for all 55 load connection points.

TABLE V
SELECTED LOCATIONAL HOSTING-CAPACITY RESULTS

Rank	Bus	HC (kWh/day)	EV-eq.	Binding
1	34.1	1127.71	47.70	Loss
2	70.1	1083.75	45.84	Loss
3	47.2	832.53	35.21	Low V
4	83.2	832.16	35.20	Low V
5	74.1	621.95	26.30	Loss
51	676.2	143.53	6.07	Low V
52	682.2	141.86	6.00	Low V
53	639.2	128.20	5.42	Low V
54	886.2	125.24	5.30	Low V
55	899.2	120.81	5.11	Low V

corresponding to 5.109 EV-equivalents, and low voltage is limiting. The strongest-to-weakest spread is approximately 9.33 times.

Fig. 9 shows substantial variation in connection-point capability. This variation confirms that a single feeder-level HC number does not capture where EV demand can be accommodated most effectively.

The Spearman rank correlation between normalized feeder

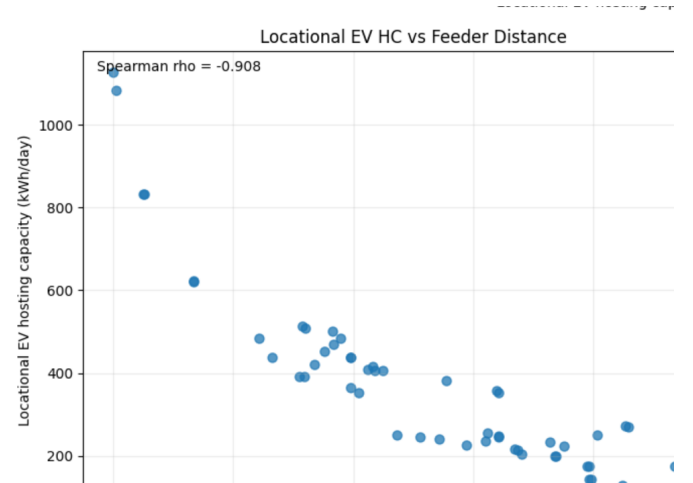


Fig. 10. Locational EV hosting capacity versus normalized feeder distance from the LV transformer.

distance and locational HC is $\rho = -0.9078$ (Fig. 10), indicating a strong negative monotonic relationship. Locations near the

transformer generally support more EV charging, while remote points become increasingly voltage-limited. This behavior is consistent with the general role of electrical distance and topology in node-level HC assessment [3], [4].

VII. DISCUSSION

The results highlight three distinct roles for physics-based HC, ML, and locational mapping. First, the unbiased Monte Carlo core quantifies natural feasibility under measured charging uncertainty. The additional transformer and loss constraints broaden the operational screening but do not artificially force a lower HC; for this feeder, low voltage remains the actual bottleneck. Second, the enriched ML datasets address class scarcity for supervised learning while remaining explicitly separated from physical HC statistics. This separation is important because an acceptance-sampled training distribution cannot be interpreted as the natural probability of feasibility. Third, the locational analysis answers a different planning question from the aggregate HC: it identifies where a standardized EV charging demand can be connected most strongly, rather than decomposing the 652.7386-kWh/day stochastic scenario.

The measured-data design also explains why the final best model differs from the reference study. Karahan and Bagriyanik [1] include synthetic bidirectional G2V/V2G behavior and select ANN for regression. The present dataset is measured, G2V-only, and represented by electrical, battery-state, spatial, temporal, and concurrency features. Under the same comparative model philosophy, RBF-SVM yields the best measured-data test performance. This model-ranking change should therefore be interpreted as a dataset-dependent result rather than evidence that one algorithm is universally superior.

VIII. LIMITATIONS AND FUTURE WORK

Several limitations should be considered. The measured EV behavior is derived from one Opel Corsa-e charging experiment; measured V2G behavior is unavailable and is therefore not synthesized. The study uses one distribution feeder and hourly simulation resolution. The 75% transformer-utilization and 2.5% loss limits are literature-based case-study thresholds from [2], not universal standards. The locational extension uses a common expected measured EV profile and continuous EV-equivalent scaling to isolate location, so its values are not literal counts of identical vehicles and are not additive across nodes.

Future work can extend the measured dataset to multiple EV models and charging powers, include measured bidirectional operation, evaluate coordinated/managed charging, test generalization on additional feeders, and investigate topology-aware ML. The node-level results also provide a natural basis for graph-based HC surrogates of the type explored in [4], while the current OpenDSS simulations remain the high-fidelity physics reference.

IX. CONCLUSION

A measured-data EV hosting-capacity framework has been developed for the IEEE European low-voltage feeder. The study combines measured SOC-dependent Opel Corsa-e charging,

CALCE OCV descriptors, 16,500 stochastic OpenDSS scenarios, a final four-constraint feasibility rule, leakage-controlled ML, and a 55-location HC extension. The unbiased core yields 1,490 feasible scenarios and a maximum observed aggregate feasible EV energy of $652.7386 \text{ kWh d}^{-1}$, with low voltage as the binding constraint. The final 14-feature SVM classifier achieves $F1 = 0.9545$, while SVM/SVR predicts daily EV energy with $R^2 = 0.9869$, $MAE = 6.5023 \text{ kWh}$, and $RMSE = 10.2364 \text{ kWh}$. The locational extension reveals a 120.81–1127.71 kWh/day range and a strong negative distance-HC relationship ($\rho = -0.9078$), demonstrating that connection location materially changes the amount of EV charging that can be accommodated. Together, the physical and data-driven results provide complementary feeder-level and node-level information for EV integration planning.

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